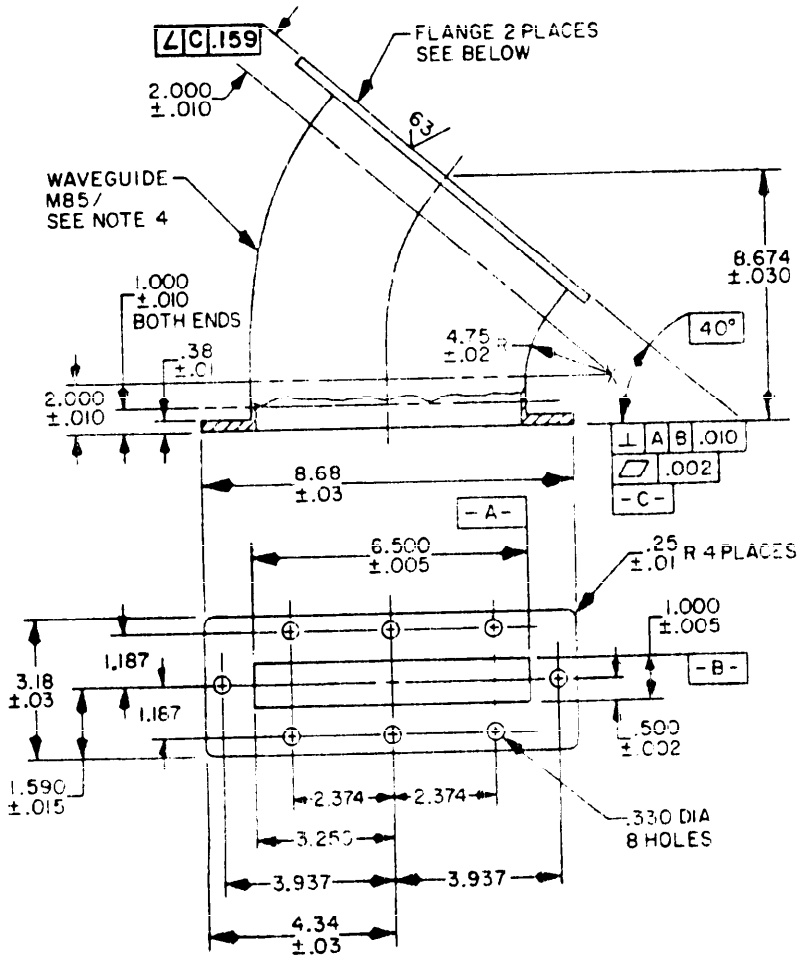
MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLY, RIGID, 40-DEGREE H-PLANE BEND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assembly described herein shall consist of this document and the latest issue of Specification MIL-W-3970.



INCHES	MM
.002	.05
.005	.13
.010	.25
.015	.38
.020	.51
.030	.76
.159	4.04
.25	6.35
.330	8.38
.38	9.65
.500	12.70
1.000	25.40
1.187	30.15
1.590	40.39
2.000	50.80
2.374	60.30
3.18	80.77
3.250	82.55
3.937	100.00
4.34	110.24
4.75	120.65
6.500	165.10
8.674	220.33
8.68	220.47

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Metric equivalents are in parentheses.
4. Waveguide per M85/1-018 except B dimension shall be 1.000 and D dimension 1.500.
5. Unless otherwise specified, tolerance is ±.002 (.05 mm).

FIGURE 1. Dimensions and configuration.

FSC 5985

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Frequency range TE_{10} mode: 1.25 - 1.35 GHz.

Voltage standing wave ratio: 1.05, maximum.

Insertion loss: .05 dB maximum.

Pressurization: 5 lb_f/in^2 gage.

Power handling capability: 350 kW peak, 2.2 kW average, at a frequency of 1.35 GHz, 50°C temperature.

Part number: M3970/20-001.

NOTE: Assemblies covered by this specification are substitutable for the following commercial types:

Part number M3970/20-	Manufacturer's code	Commercial type
001	49956	712068
001	23572	712068

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, EL
Navy - EC, OS
Air Force - 11, 17, 99
DSA - ES

User activities:
Army - MU
Navy - MC, CG, AS
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5985-0835-17)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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